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Roll No

CE-501 (GS)
B.E. V Semester Examination, June 2020
Grading System (GS)
Transportation Engineering - II
Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Discuss salient features of Nagpur Road plan.
b) Derive an expression for finding superelevation on highway.
2. A vertical summit curve is formed at the intersection of two gradients, + 3.0 and – 5.0 percent. Design the length of summit curve to provide a stopping sight distance for a design speed of 75 kmph. Assume $t = 2.5$ sec and $f = 0.30$
3. How is the alignment of highway selected through various considerations?
4. Explain the different tests carried on Bitumen in detail with the help of neat figure
5. a) Calculate the ESWL for a wheel load of 2050kg in a dual wheel axle to be applied on a road of 300mm thick. The distance between centre of wheel to be taken as 250mm and the edge distance between types as 120mm.
b) What are the requirements of good joints? Why are joints provided in cement concrete pavement.
6. What is Soil Stabilization? How it is carried out by different methods? Explain in detail.

OR

What do you understand by the term “runway capacity”? What are the factors affecting the runway capacity?

7. a) What is zoning? Explain zoning regulations.
b) Explain the necessity of airport lighting.

OR

- a) Differentiate between channelized and unchannelized intersections.
- b) Distinguish between WBM, WMM, BM & IBM.
8. Explain the following:(any two)
 - a) Super Elevation
 - b) Sight distances
 - c) Extra widening
 - d) Camber
